

Carnegie Mellon University

Case Study: LendingClub Diagnostic & Descriptive Analytics

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Business Understanding & KPIs

- Re-allocation
 - Notes with better risk-adjusted return
- Adhere to diversification / risk limits
- KPIs
 - Annualized return
 - Default rate
 - Churn / loss rate
 - Grade / purpose mix
 - Baseline
 - Model lift vs. grade-only

Fairness & Limitations

- Data Limitations
 - ~4.8% average missingness per dataset & data is a decade old
 - Survivorship bias since data only covers loans that reached an outcome
 - Reduction of transparency for stakeholders due to PCA
- Fairness Considerations
 - Protected attributes aren't observed, but proxy risk exists
 - Risk of disparate impact
 - Clustering or returns optimization could overweight loans from certain groups
 - Ethical modeling with business KPIs
 - Equal Credit Opportunity Act

Business Understanding & KPIs

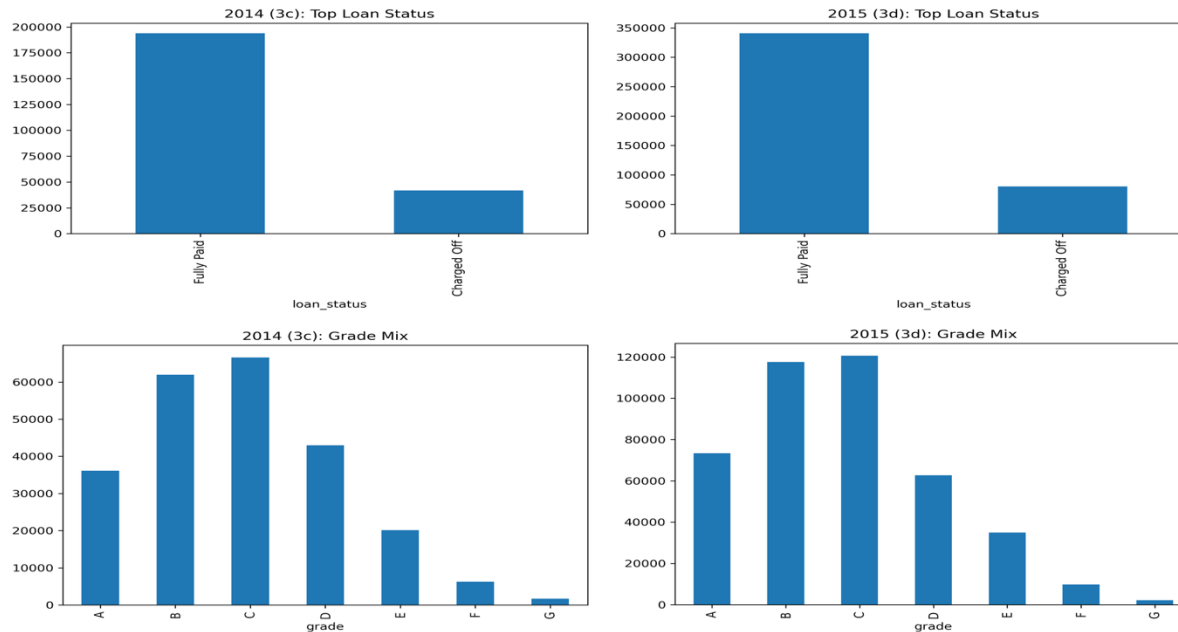


Figure 01. Baseline cohort summaries. Top row: 2014 & 2015 loan status breakdown .
Bottom row: grade distribution for 2014 and 2015.
Bars show counts and y-axes differ by panel.

Data Cleaning & Preprocessing

Standardization

- Converted rates from percent to float and documented ~4.8% average missingness per dataset
- Parsed 36 months as terms and parsed dates to datetime

Normalization

- Employment length mapped to numeric bins
- Fixed category dtype ('grade', 'verification_status', 'purpose', etc.)

Leakage Prevention

- Dropped post-outcome fields from clustering and PCA inputs
- Kept only for returns computation

Notes

- Ensures features are comparable and reproducible
- Prevents info leakage into diagnostics or modeling

Cohen Return Scenarios

Scenario	Possible Solution
Optimistic	Use total payment; including recoveries, excluding service fees
Intermediate	Subtract ~1% servicing on total payments
Pessimistic	For defaults, only recoveries; fully paid as Intermediate
Note	Report medians by grade and term. Note proxy vs. IRR

$$\text{ROI} = (\text{total payment} - \text{funded amount}) / \text{funded amount}$$

$$\text{Annualized} = (1 + \text{ROI})^{(12/\text{term})} - 1$$

Cohen Return Scenarios

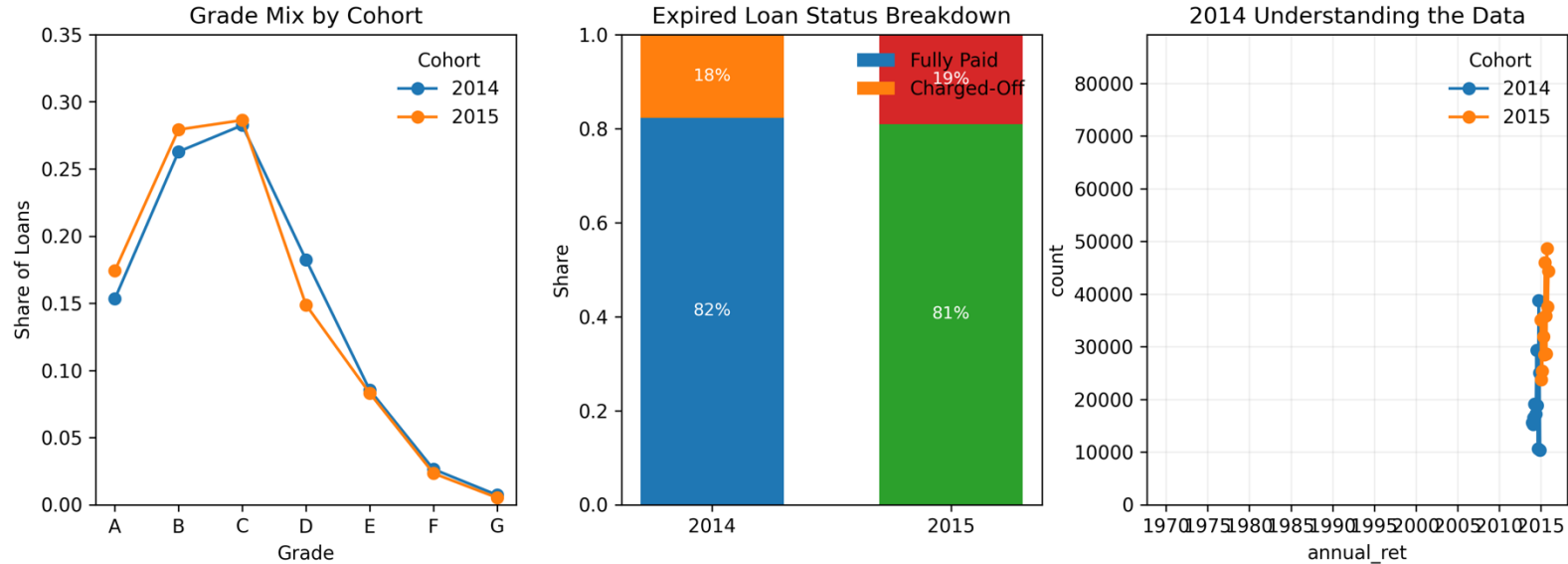


Figure 02. 2014 & 2015 Grade distribution by cohort, loan status breakdown, and annualized return distribution.

Cohen Return Scenarios

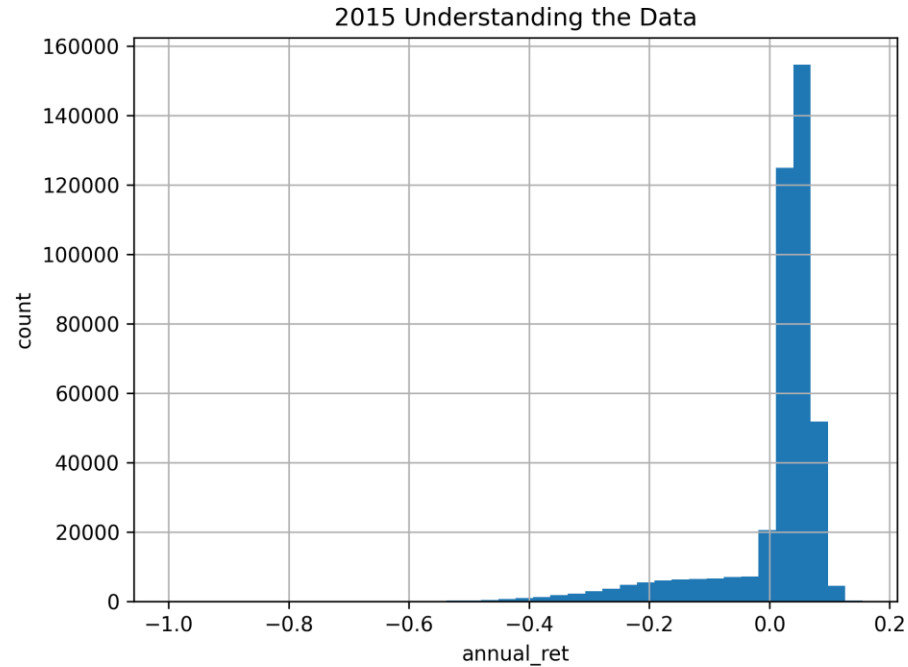


Figure 03. 2015 LendingClub loans Distribution of annualized returns

Cohen Return Scenarios

	loan_status	roi_opt	roi_pess	roi_int
0	Fully Paid	0.159491	0.159491	0.159491
1	Charged Off	-0.364261	-0.364261	-0.364261
2	Charged Off	-0.701702	-0.701702	-0.701702
3	Fully Paid	0.497293	0.497293	0.497293
4	Fully Paid	0.190768	0.190768	0.190768

Figure 04. Example loan statuses with ROI computed under optimistic (M1), pessimistic (M2), intermediate (M3) scenarios.

Cohen Return Scenarios

	grade	term_months	N	m1_mean	m1_median	m2_mean	m2_median	m3_mean	m3_median
0	A	36.0	70132	0.016382	0.027445	0.014126	0.027421	0.014522	0.026558
1	A	60.0	3204	0.019119	0.031166	0.016461	0.031150	0.017502	0.030634
2	B	36.0	91783	0.015270	0.038553	0.010097	0.038511	0.012092	0.037637
3	B	60.0	25823	0.017307	0.037594	0.011976	0.037538	0.014502	0.037019
4	C	36.0	77457	0.008407	0.051946	-0.001032	0.051783	0.003331	0.050921
5	C	60.0	43110	0.013212	0.045137	0.004541	0.044796	0.008885	0.044349
6	D	36.0	32740	0.000054	0.057634	-0.014127	0.056978	-0.007029	0.056229
7	D	60.0	29914	0.003366	0.044995	-0.009827	0.043861	-0.002886	0.043634
8	E	36.0	9450	-0.012082	0.056109	-0.031130	0.054816	-0.021213	0.054393
9	E	60.0	25498	-0.004984	0.036115	-0.022078	0.033790	-0.012792	0.033871
10	F	36.0	1363	-0.028676	0.044708	-0.055756	0.039617	-0.040997	0.041073
11	F	60.0	8454	-0.018824	0.023386	-0.041852	0.018322	-0.028846	0.019518
12	G	36.0	248	-0.044272	0.035647	-0.082135	0.032678	-0.060865	0.031808
13	G	60.0	1919	-0.031293	0.013892	-0.058548	0.008079	-0.042955	0.009029

Figure 05. Mean and median returns (M1, M2, M3) by loan grade and term length.

2014 K-Means Clusters

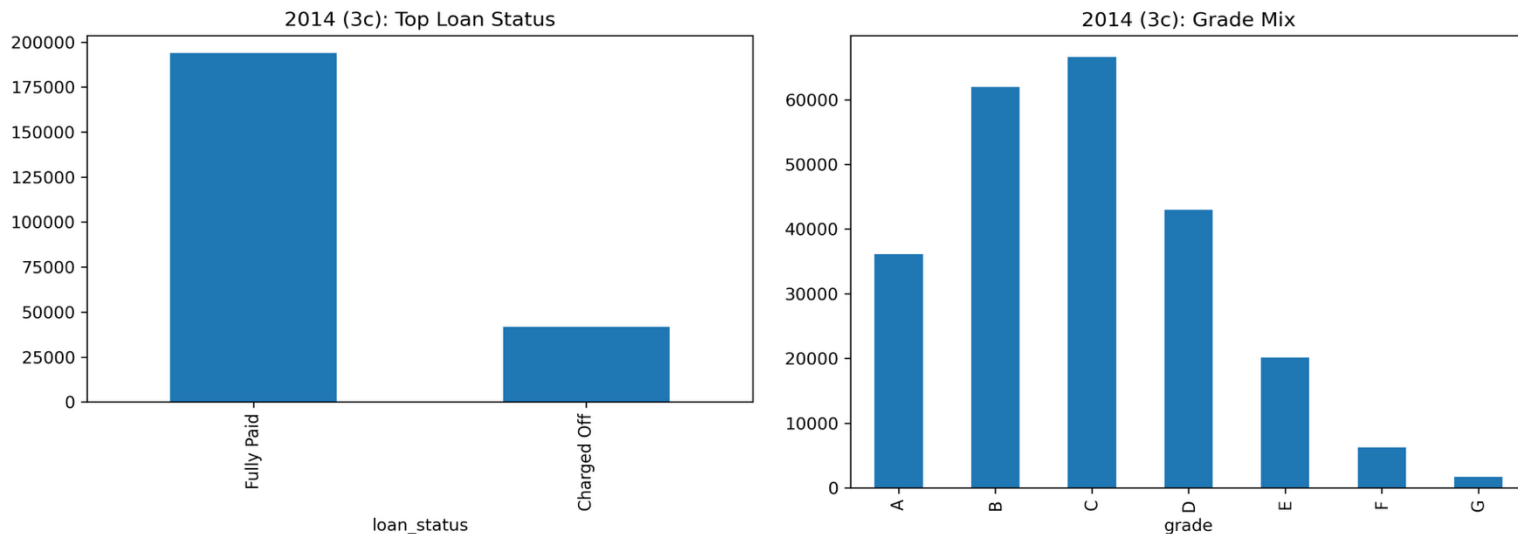


Figure 06. Standardized + OHE with sweep `k = 3.8`. Chosen by silhouette + interpretability.

2014 silhouette by k: {3: 0.12935200407407615, 4: 0.08577847767598296, 5: 0.0896640479567341, 6: 0.09106730609964137, 7: 0.08772625523489137, 8: 0.09012480509547821}

2015 K-Means Clusters

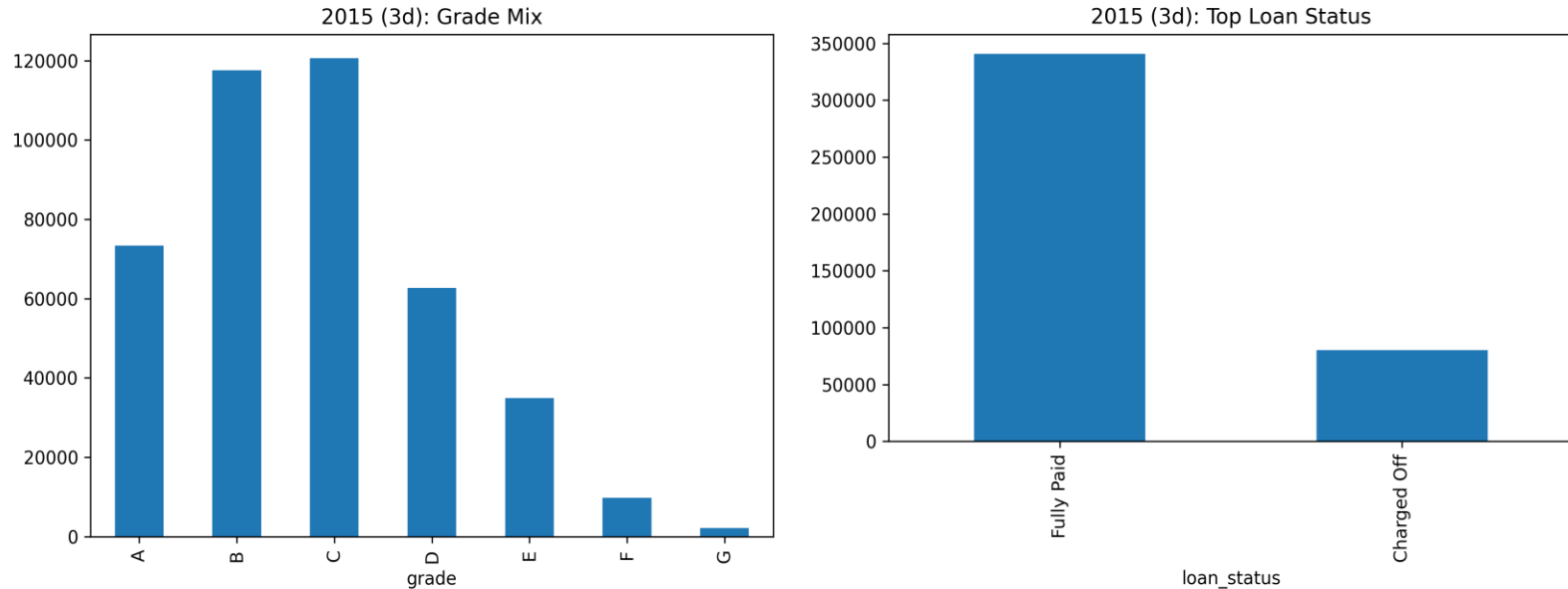


Figure 07. 2015 silhouette by k: {3: 0.12695837041556549, 4: 0.08368638298390696, 5: 0.08717727962986599, 6: 0.08842193608715984, 7: 0.08864946347798841, 8: 0.0914693515173938}

Cluster Scatterplots (2D)

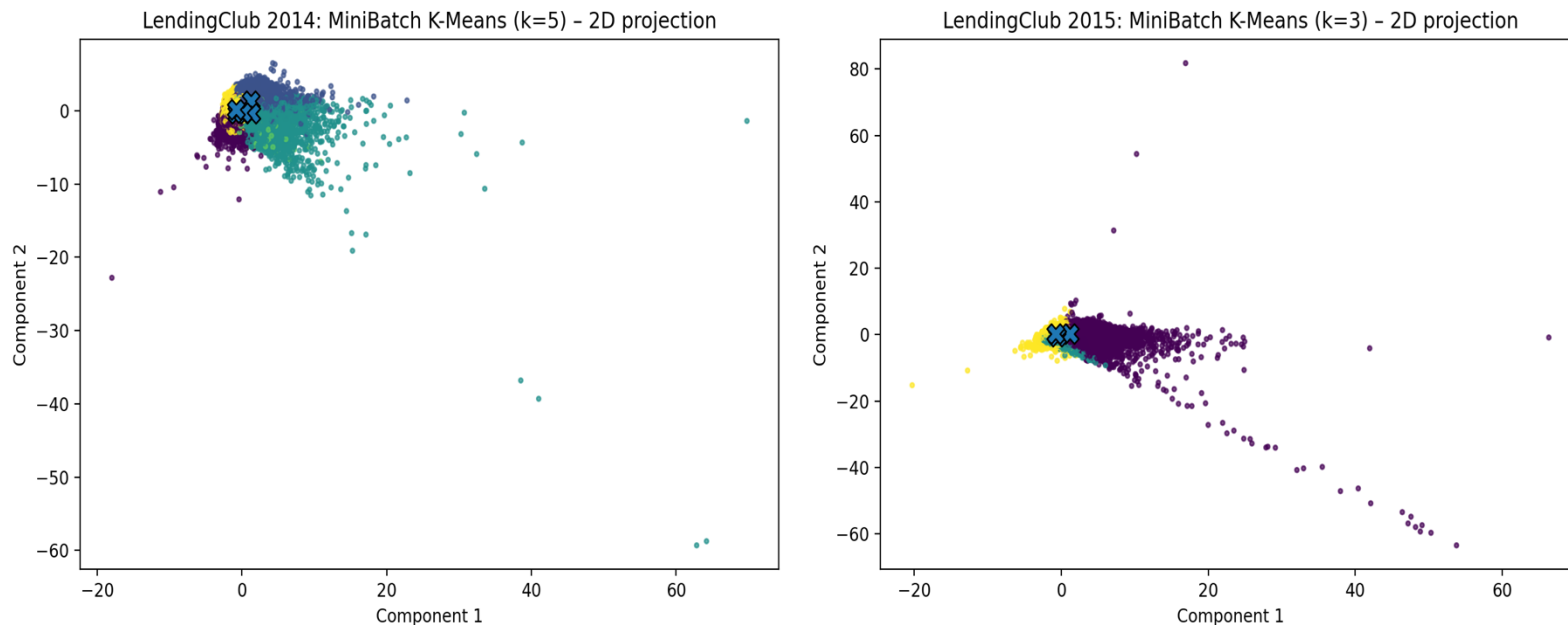


Figure 08. 2014 & 2015 LendingClub MiniBatch K-Means 2D projection ($k = 5$).

Silhouette by k

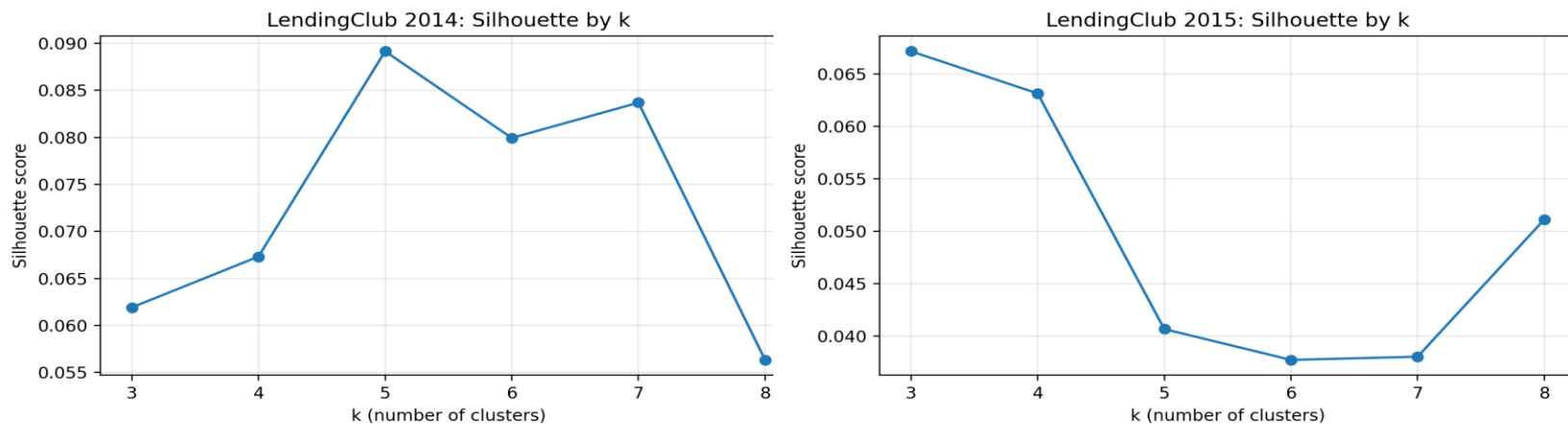


Figure 09. 2014 & 2015 LendingClub silhouette scores by k.

Cluster Profiles

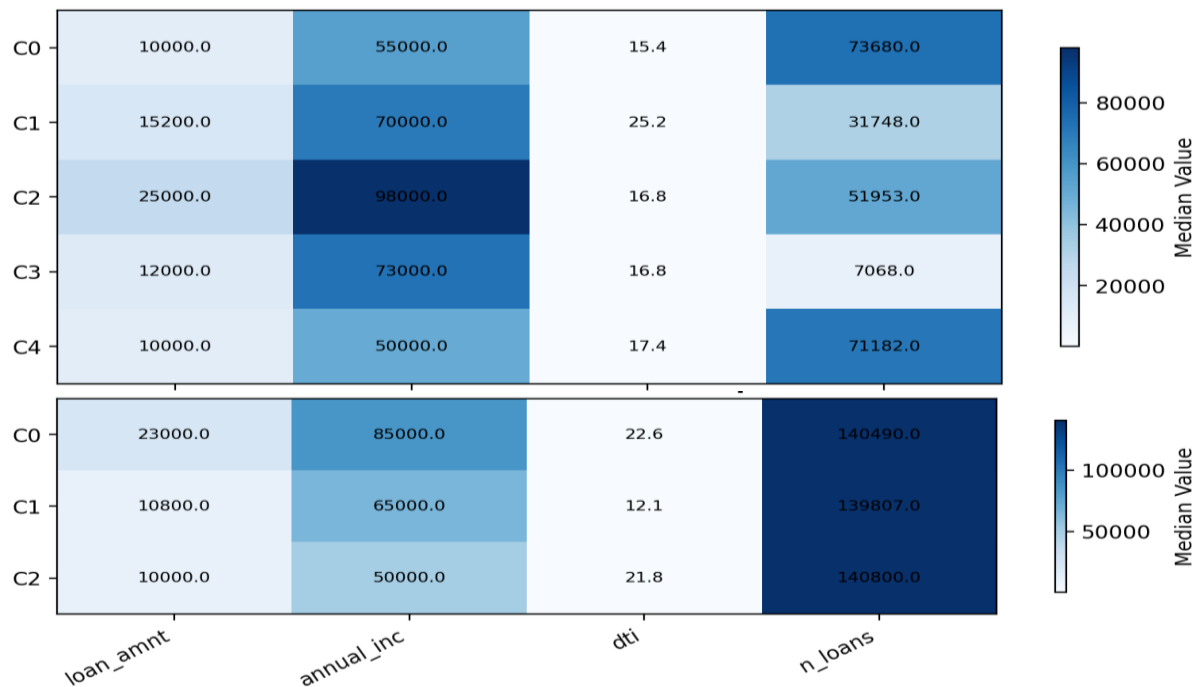


Figure 10. Heatmaps of LendingClub cohort median cluster characteristics.
Top: 2014. Bottom: 2015.

Cluster Profiles

	loan_amnt	annual_inc	dti	n_loans
C0	10000.0	55000.0	15.4	73680.0
C1	15200.0	70000.0	25.22	31748.0
C2	25000.0	98000.0	16.8	51953.0
C3	12000.0	73000.0	16.8	7068.0
C4	10000.0	50000.0	17.45	71182.0

	loan_amnt	annual_inc	dti	n_loans
C0	23000.0	85000.0	22.65	140490.0
C1	10800.0	65000.0	12.1	139807.0
C2	10000.0	50000.0	21.78	140800.0

Figure 11. Cluster Profiles Table. Top: 2014. Bottom: 2015.

K-Means Cluster Sizes & Median Annual Ret

2014 Cluster Sizes		2015 Cluster Sizes		median_annual_ret	
cluster		cluster		cluster	
0	69209	0	127464	0	0.047819
1	40041	1	228020	1	0.040737
2	126379	2	65611	2	0.045215
default_rate		default_rate		median_annual_ret	
cluster		cluster		cluster	
0	0.276510	0	0.294287	0	0.041792
1	0.106766	1	0.154013	1	0.039971
2	0.145087	2	0.116916	2	0.036912

Figure 12. 2014 & 2015 Cluster Sizes and respective median annual returns.

PCA Cluster vs. Grade

2014 Cluster vs Grade Distribution (row-norm)							
grade	A	B	C	D	E	F	G
cluster							
0	0.004826	0.092228	0.301218	0.321259	0.192981	0.065107	0.022381
1	0.320921	0.330661	0.231987	0.088359	0.023551	0.003646	0.000874
2	0.181391	0.334802	0.288252	0.136257	0.046068	0.012431	0.000799
2015 Cluster vs Grade Distribution (row-norm)							
grade	A	B	C	D	E	F	G
cluster							
0	0.010983	0.142817	0.308040	0.246156	0.207847	0.068278	0.015879
1	0.218152	0.336199	0.292286	0.117520	0.031067	0.004223	0.000553
2	0.338251	0.346619	0.223377	0.068296	0.020896	0.002301	0.000259

Figure 13. Cluster vs. Grade Distribution (row-normalized)

PCA Parallel-Coordinates

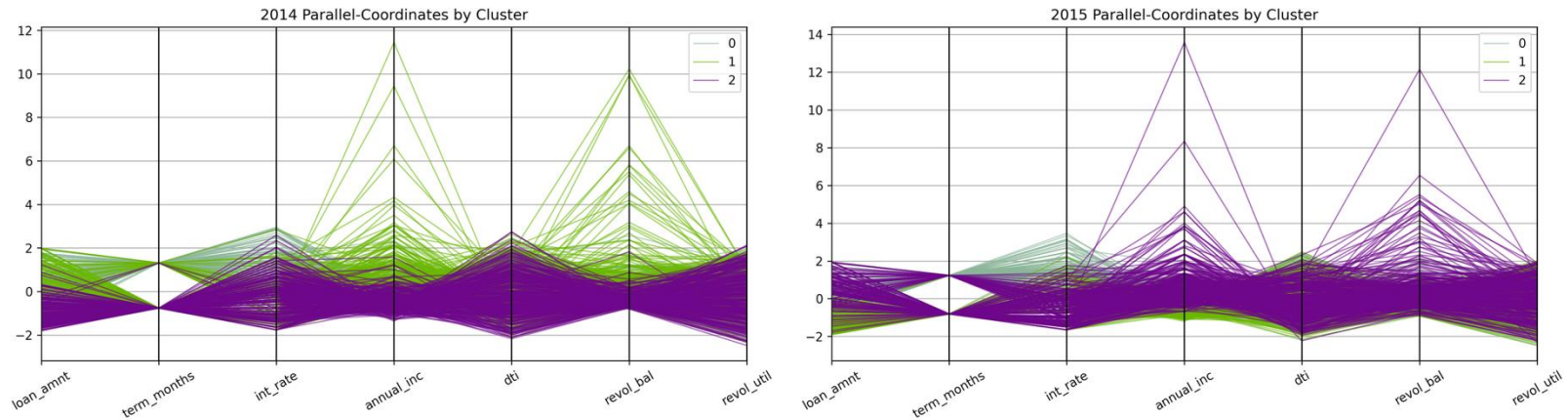


Figure 14. Parallel-coordinates of standardized key features by cluster ($k = 3$). Each line is a loan and color indicates cluster. Left: 2014. Right: 2015.

Cluster Prototypes

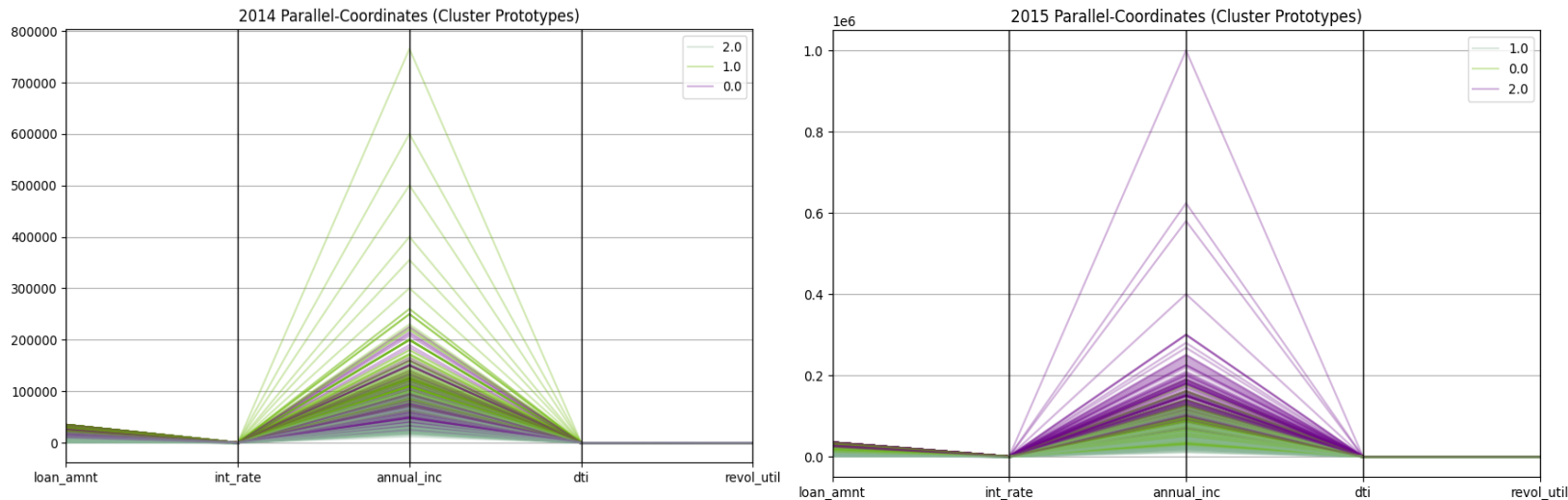


Figure 15. Cluster prototypes by median feature profiles visualized with parallel-coordinates. Left: 2014. Right: 2015.

SMOTE & imblearn

Let default=1. Then,

- 2014 minority share = 0.177,
- 2015 minority share = 0.191,

and when SMOTE is demonstrated on a tiny numeric sample,

- Before = {0:193881, 1:41748},
- Before = {0:193881, 1:193881}.

Future Directions

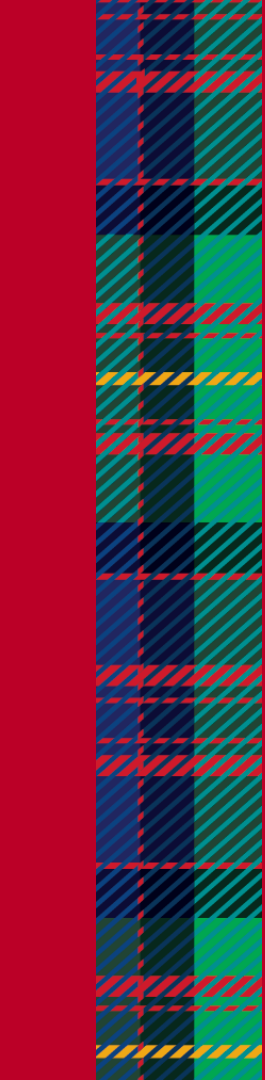
Goals

Calibrated Default Model + Backtesting by Issue Date

Investment Policy

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Thank You!

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